

Semantic Retrieval with Sentence-Transformers and Firestore Vector Storage

Abstract

This paper presents a novel approach to semantic retrieval using sentence-transformers embeddings and Firestore vector storage. We demonstrate that a 384-dimensional embedding model combined with persistent vector storage can achieve effective semantic search for research papers.

Introduction

Research in natural language processing has made significant advances in document understanding and retrieval. Traditional keyword-based approaches are limited in their ability to capture semantic meaning. This paper proposes a hybrid approach that combines modern embedding techniques with persistent vector storage.

Methodology

Our approach uses the all-MiniLM-L6-v2 sentence-transformers model to generate 384-dimensional embeddings for document chunks. These embeddings are stored in Firestore as vector documents with metadata including `workspace_id`, `paper_id`, and `chunk_id`.

Embedding Generation

We use the sentence-transformers library to generate embeddings for each document chunk. The embedding model is configured to output 384-dimensional vectors that capture semantic meaning of the text.

Vector Storage

Vectors are stored in Firestore's `workspace_vectors` collection with the following structure: `embedding` (384-dimensional float array), `workspace_id` (string identifying the workspace), `paper_id` (string identifying the paper), `chunk_id` (string identifying the specific chunk), and `text` (the original text content).

Similarity Calculation

We use cosine similarity to measure the semantic similarity between query embeddings and stored document vectors. A similarity threshold of 0.2 is used to filter relevant results.

Results

Our experiments show that the proposed approach achieves effective semantic retrieval with high precision in retrieving relevant document chunks, low false positive rate, good performance on multi-chunk queries, and persistent storage enabling retrieval after system restarts.

Conclusion

The combination of sentence-transformers embeddings and Firestore vector storage provides an effective solution for semantic document retrieval. The 384-dimensional embeddings capture sufficient semantic information while maintaining computational efficiency.

Future Work

Future research directions include exploring larger embedding models, optimizing similarity thresholds, implementing multi-modal retrieval, and adding citation verification.